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# Feasibility study for a standalone solar–wind-based hybrid energy system for application in Ethiopia

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ABSTRACT

The aim of this paper is to investigate the possibility of supplying electricity from a solar–wind hybrid system to a remotely located model community detached from the main electricity grid in Ethiopia. The wind energy potential of four typical locations has been assessed in a previous article. The solar potential has also been investigated and the results are presented in detail in an accompanying article awaiting publication. For one of the sites, Addis Ababa, the results of the investigation are given here in detail. For the other sites, the results are given as sensitivity diagrams only. Based on the findings of the studies into energy potential, a feasibility study has been carried out on how to supply electricity to a model community of 200 families, which comprises 1000 people in total. The community is equipped with a community school and a health post. The electric load consists of both primary and deferrable types and comprises lighting, water pumps, radio receivers, and some clinical equipment. A software tool, Hybrid Optimization Model for Electric Renewables (HOMER) is used for the analysis. The result of the analysis is a list of feasible power supply systems, sorted according to their net present cost. Furthermore, sensitivity diagrams, showing the influence of wind speeds, PV costs, and diesel prices on the optimum solutions are also provided.



## 1. Background

The enormous problems in Ethiopia caused by the shortage of power have been outlined in a previous article [1]. The recurrent droughts, with which the country has long since been associated, and the adversity which has continued unabated, even into the current century, can be said to be directly and/or indirectly attributed to the shortage of power. As recently as 2008 drought afflicted the country and millions needed food aid.

The heavy reliance of the population on biomass to meet their urban and rural energy needs has depleted the forest, and the use of crop residues and dung for fuel has resulted in centuries of cropland degradation. The seriousness of the land degradation, which is increasingly undermining the agriculture that the majority of the population depends on, has put the country in a quagmire of food insecurity. Paradoxically, it is the still diminishing biomass stock which continues to be the main source of energy in most parts of the country.

The resettlement program which the government is pursuing in an effort to alleviate the problem has been mentioned in a previous article [1]. Over a million people have already been moved to different parts of the country, where land conditions are better although even here there is no electricity.

The Ethiopian Electric Power Corporation (EEPCo), the sole electric power producer in the country, currently generates much less than 1000 MW of power. To be precise it was 2895 GW h of electricity in the year 2005/2006. This is for a country with a population of approximately 75 million and where the electricity generated comes mainly from a few major hydro-power plants. Other self-contained sources, i.e., those not connected to the grid, such as geothermal (steam), mini hydro-power plants and a number of isolated diesel generators also contribute to the total sum of less than 1000 MW. From this basic fact alone, it is clear to see what an alarming situation the country is in.

As most areas of the country are not connected to the electric grid, and as the power to the grid is insufficient, rural areas are dependent on local solutions for electricity supply. The standard solution has traditionally been to use diesel generators for this supply. However, price increases in imported oil, which are anticipated for the near future, and the negative effects of fossil fuels on the local and global environment motivates the search for other alternatives.

When looking for alternative resources it is solar and wind energy that are the first options on the list. These resources are locally available and they are free, in addition to being environment friendly.

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## 2. Wind energy potential

The wind energy potential of the country is not particularly extensive, as the country has no coastline. However, due to its geographical location, the summer monsoon, tropical easterlies, and local convergence over the Red Sea do play a significant role in the windiness of the country [3]. The topography itself also contributes a significant portion.

Based on data obtained from the Ethiopian National Meteorological Services Agency (NMSA), a report by Deutsche Gesellschaft für Technische Zusammenarbeit (GTZ) [4], software and satellite data obtained from Meteonorm [5] and NASA [6], and data from the website Weatherbase[SM] [7], the average monthly wind speed of four locations in the country was determined with the help of a software tool [8]. The four locations are Addis Ababa, 09″02′N, 38″42′E, 2408 m (AMSL); Mekele, 13″33′N, 39″30′E, 2130 m; Nazret 08″32′N, 39″22′E, 1690 m; and Debrezeit, 08″44′N, 39″02′E 1850 m. The findings, reported previously [1], indicate that wind power is worthy of consideration for standalone power supply systems. Fig. 1 shows the results in terms of monthly average wind speeds.

## 3. Solar energy potential

Statistical data shows that tropical regions offer a better solar energy resource than at more temperate latitudes. The average annual irradiation in Europe is about 1000 kW h/m$^2$, while in the middle east the value is approximately 1800 kW h/m$^2$. In the tropical zone, of which Ethiopia is a part, the average is estimated to be around 2000 kW h/m$^2$.

Of the four locations investigated here, properly recorded solar radiation data is available only for Addis Ababa. The radiation at ground level at the other sites is estimated in our accompanying article using sunshine duration data and appropriate empirical relationships. The sunshine duration data used in the simulation is based on data recorded for more than 10 years, for which the source is the NMSA. The Meteonorm Global Meteorological Database [5] and the renewable energy resource web site, sponsored by NASA [6] are the other sources used in the process of the investigation [2].

Fig. 2 shows the monthly average daily solar radiation of the four locations. More information about the determination of this data is found in the accompanying paper [2].

## 4. Electrical load

Deciding on the load is one of the most important steps in the design of a hybrid system. In the current study, a hypothetical model community of 200 families, each comprising of five family members, is considered. A community school and a health post are also provided for the community. The load is of two types, a primary load, i.e., a load that must be met immediately, and a deferrable load, i.e., a load that must be met within a certain time frame (although the exact timing is not important). The primary electric load consists of lighting, radio receivers, and some clinical equipment.

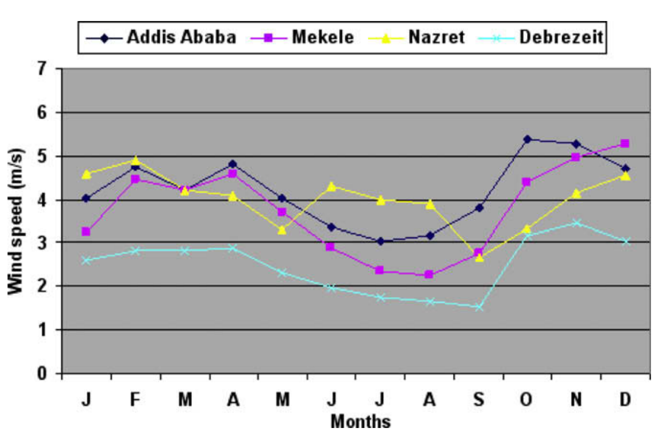



Fig. 1. Monthly average wind speeds at the four investigated locations.

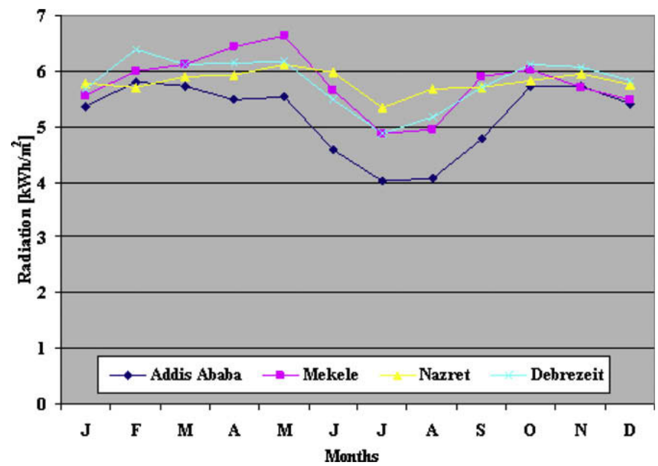



Fig. 2. Monthly average daily solar radiation.

The deferrable load incorporates six water pumps for the households and another for the school and health post, each with a 150 W power rating and a pumping capacity of 10 l/min. These are able to pump 20,000 l/day for the 200 families (100 l per family); and 2400 l/day for the school and the clinic. The average deferrable load is therefore 5.4 kW h for the households and 0.6 kW h for the school and health post, which totals 6 kW h. The peak deferrable load is 0.9 kW for the households and 0.15 kW for the school and health post. Assuming a water storage capacity of 4 days, the corresponding electricity storage capacity is 20 kW h for the households and 2.4 kW h for the school and health post.

The peak primary load per household is assumed to be 130 W (5 W night light, 3 W radio receiver and two 60 W light bulbs). The total daily consumption of the families is thus assumed to be 138 kW h. The principal component of this consumption is 5 h of lighting each night.

Electric lighting is proposed for the school for those who wish to pursue basic education in the evenings (18:00–21:00). Considering the 17 necessary light sources, each with a capacity of 40 W, the total consumption would be 2.0 kW h/day.

A typical two-room health care facility, equipped with vaccine refrigerator, light bulbs, Stand-by Communication VHF radio, microscope, and AM/FM radio is also proposed.

The assumption here is that the "health post" would not be a permanently-staffed type, providing constant health services with a permanent doctor or nurse present; but, rather a doctor or a nurse would periodically provide treatment for minor illnesses and tend to minor injuries. Patients with more serious problems would be referred to the nearest hospital. For this reason only the most basic items are considered. The daily consumption is thus calculated to be approximately 1 kW h.

The total daily energy consumption for the community of 200 families is therefore the sum total of household, school and health post consumption, plus the deferrable load, which adds up to 147 kW h.

Some minor exceptions to the aforementioned value should be considered. July and August are the rainy season in the country and schools are closed; as is also the case for January, which is a semes-

**Table 1**
Monthly average daily electrical load in kW h.

| | Months | | | | | | |
|---|---|---|---|---|---|---|---|
| | January | February–May | June | July | August | September | October–December |
| Deferrable load | 6 | 6 | 5 | 4 | 4 | 5 | 6 |
| Primary load | 139 | 141 | 141 | 139 | 139 | 141 | 141 |
| Total load | 145 | 147 | 146 | 143 | 143 | 146 | 147 |

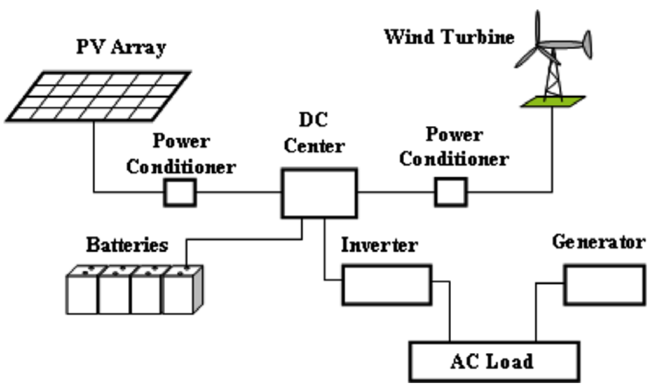



**Fig. 3.** General scheme for the standalone hybrid power supply system.

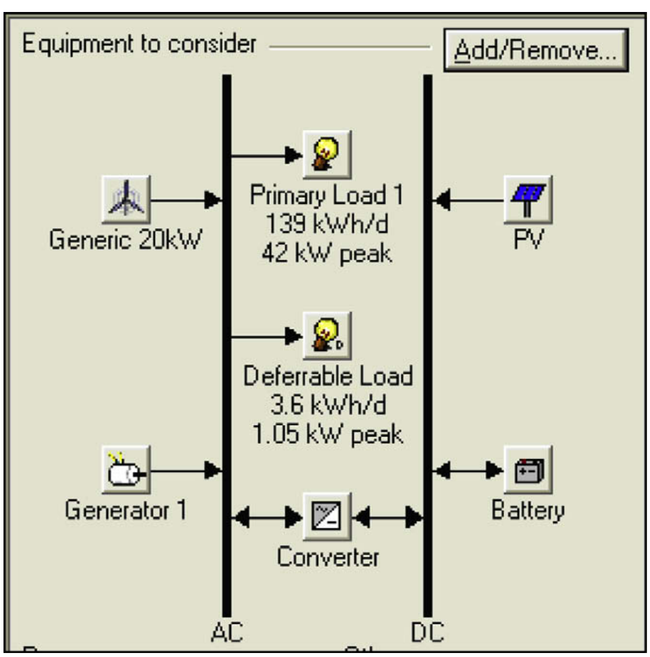



**Fig. 4.** HOMER diagram for the hybrid PV–wind–generator–battery–converter setup.

ter break. During the rainy season water consumption from the pumps is supposed to be supplemented by rain water ponds or river water and if this is assumed to account for a 30% share of the pump load, the load in these months becomes approximately 140 kW h and about 145 kW h in January. Hence a typical load pattern over the course of a year is given in Table 1.

The load calculated thus far is constant for each day of the month. In reality, the size and shape of the load profile will vary from hour to hour and from day to day. Hence, on a daily and hourly basis a 15% noise level has been added to the calculated load in order to randomize the load profile and make it more realistic. This has scaled up the annual peak load to 42 kW, as can be observed in Fig. 4.

Other information which has been input to the calculation program is summarized in Table 2. This information includes the sizes and prices of the hybrid setup components which have been obtained from the respective vendors [9].

## 5. The hybrid system and the setup

As previously shown [2] there is abundant solar energy, with a daily average amounting to some 6 kW h/m$^2$. Integrated with other components, such as PV, wind energy could also be used for a standalone power supply system. These resources can be utilized for electric power generation. However, the question as usual concerns the cost. The capital cost of PV systems has always been the main barrier to the use of solar energy and also to the promotion of PV technology for large-scale applications. The current crystalline silicon type PV capital cost, i.e., the total installed cost of PV at the beginning of the project, is assumed to be 4000 $/kW, while that of conventional oil- or gas-based power systems is approximately 1000 $/kW [10]. Despite its high initial capital cost, for a country like Ethiopia where there is only 15% electricity grid coverage, PV systems could also be competitive, especially when considering the rapid increase in the price of oil. It should also be noted that there are advantages associated with the use of PV, such as a minimal maintenance cost, long lifetime and the possibility of easily expanding the systems. The negligible impact of PV systems on the global and local environments is also an important advantage. Nonetheless, with advances in technology, investment in production facilities and increasing fuel prices, solar power is becoming a serious candidate in the electricity market. According to a joint report launched by the European Photovoltaic Industry

**Table 2**
Summary of software inputs.

| Item | Size | Capital ($) | Replacement cost ($) | Operation and maintenance cost ($/year) | Sizes (kW) considered | Quantities considered | Life time |
|---|---|---|---|---|---|---|---|
| PV | 1 | 4000 | 4000 | 0 | 0–100 | | 25 years |
| AC wind turbine (generic type) | 20 | 45,000 | 30,000 | 900 | | 0–3 | 25 years |
| AC generator | 44 | 11,000 | 7000 | 0.4 $/h | 0, 44, 88 | | 40,000 h |
| Battery (surrette 6CS25P) 9645 kW h | 1156 Ah | 833 | 555 | 15.00 | | 0–200 | |
| Converter | 1 | 700 | 700 | 0 | 0–100 | | 15 years |

Association (EPIA) and Greenpeace [11], two billion households worldwide could realistically be powered by solar energy by 2025.

However, PV alone cannot supply energy on a 24 h basis and therefore has to be supported by alternative energy sources such as wind and/or Genset, batteries and others. Wind power provides obvious advantages, but to be truly competitive with conventional energy sources, it must also be economical. The cost of a wind system with operational battery storage in a remote location is estimated to be between 4000 and 5000 \$/kW [12]. This cost is about the same as that of PV, but in light of the reasons given above, and when other alternatives, such as hydro power, are not available close to the site, or an extension of the gridline is comparatively expensive, the use of the wind energy becomes a realistic consideration. A lack of reliability in supply, however, remains a problem with wind energy too, as the wind does not always blow. The variation in energy production of both solar and wind resources mismatches, at least partially, with the time distribution of the load demand. Hence, power generation from solar and wind systems dictate the associated use of diesel generators and/or battery storage facilities in order to ensure a constant power supply. The battery bank stores energy when excess wind and solar energy is available and releases it when needed.

In this study the main objective is to assess the feasibility and economic viability of utilizing hybrid PV–wind–diesel–battery power systems to meet the load requirements of a typical hypothetical community of 200 families.

The schematic diagram of the standalone hybrid power supply system under investigation is shown in Fig. 3 and its representation by HOMER is also shown in Fig. 4.

## 6. Results and discussion

As part of the project literature survey is carried out and findings published by several researchers have been investigated, special attention being given to the most recently published results. It is well known that researchers have been working diligently to optimize solar–wind-based hybrid electricity supply systems for different applications. The type of application may vary from small scale supply system, such as for energizing a simple Global System Communication (GSM) base station or a remote consumer [13–18] to relatively larger scale such as rural electrification for larger community [19]. In the optimization process different types of simulation models, such as ARENA, commercial software and Opt-Quest tool [15,16], Matlab/Simulink [19], response surface metamodels (RSM) [13], genetic algorithm (GA) [14] have been used. Based on a variety of design parameters; such as PV size, wind turbine rotor swept area, battery capacity, PV module slope angle, and wind turbine installation height levelized cost of energy (\$/kW h) ranging in between \$1 and \$4 have been estimated. This cost compared to the current global electricity tariff (conventional electric power generation system), which is ranging in between 3.56 in South Africa and 36.74 US cents/kW h in Italy [20] is very high.

**Table 3**
Overall optimization results according to net present cost (NPC).

| PV (kW) | G20 | Genl (kW) | Battery | Converter (kW) | Display strategy | Initial capital (\$) | Total NPC (\$) | COE (\$/kW h) | Renewable fraction | Diesel (l) | Gen (h) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 44 | 40 | 20 | CC | 58,320 | 201,609 | 0.322 | 0 | 18,623 | 1785 |
| 5 | | 44 | 40 | 20 | LF | 76,320 | 220,728 | 0.353 | 0.16 | 18,115 | 2391 |
| | | 44 | 40 | 20 | LF | 58,320 | 222,616 | 0.356 | 0 | 21,056 | 2725 |
| 10 | | 44 | 40 | 20 | LF | 94,320 | 226,668 | 0.362 | 0.3 | 16,217 | 2293 |
| | | 44 | 60 | 20 | CC | 74,980 | 227,227 | 0.363 | 0 | 18,605 | 1788 |
| 5 | | 44 | 60 | 20 | CC | 92,980 | 230,456 | 0.369 | 0.15 | 16,320 | 1638 |
| 15 | | 44 | 40 | 20 | LF | 112,320 | 231,855 | 0.371 | 0.43 | 14,260 | 2137 |
| | 1 | 44 | 40 | 20 | LF | 103,320 | 233,435 | 0.373 | 0.39 | 14,355 | 2021 |
| 5 | | 44 | 40 | 40 | LF | 90,320 | 238,842 | 0.382 | 0.16 | 18,083 | 2379 |
| 5 | 1 | 44 | 40 | 20 | LF | 121,320 | 239,756 | 0.383 | 0.51 | 12,599 | 1858 |
| 20 | 1 | 44 | 60 | 40 | LF | 205,980 | 289,942 | 0.464 | 0.81 | 5662 | 817 |
| 10 | 2 | 44 | 40 | 20 | LF | 184,320 | 290,411 | 0.464 | 0.75 | 9177 | 1460 |
| 30 | | 44 | 80 | 40 | LF | 213,640 | 300,698 | 0.481 | 0.77 | 6379 | 914 |
| 10 | 1 | 44 | 80 | 40 | LF | 186,640 | 300,866 | 0.481 | 0.65 | 8812 | 1222 |
| | | 44 | 80 | 40 | CC | 105,640 | 300,887 | 0.481 | 0 | 23,005 | 2243 |
| | | 88 | 40 | 40 | LF | 83,320 | 300,892 | 0.481 | 0 | 27,598 | 2294 |
| 15 | | 44 | 40 | 100 | LF | 168,320 | 300,932 | 0.481 | 0.43 | 13,767 | 1919 |
| 15 | 1 | 44 | 60 | 60 | LF | 201,980 | 300,959 | 0.481 | 0.73 | 7239 | 1022 |
| 5 | | 44 | 60 | 80 | LF | 134,980 | 301,094 | 0.481 | 0.16 | 18,051 | 2374 |
| 5 | | 44 | 40 | 60 | CC | 104,320 | 301,644 | 0.482 | 0.13 | 24,133 | 3577 |

**Table 4**
Optimization results, in a categorized form, ranked according to the NPC of each system type.

| PV (kW) | G20 | Genl (kW) | Battery | Converter (kW) | Display strategy | Initial capital (\$) | Total NPC (\$) | COE (\$/kW h) | Renewable fraction | Diesel (l) | Gen (h) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 44 | 40 | 20 | CC | 58,320 | 201,609 | 0.322 | 0 | 18,623 | 1785 |
| 5 | | 44 | 40 | 20 | LF | 76,320 | 220,728 | 0.353 | 0.16 | 18,115 | 2391 |
| | 1 | 44 | 40 | 20 | LF | 103,320 | 233,435 | 0.373 | 0.39 | 14,355 | 2021 |
| 5 | 1 | 44 | 40 | 20 | LF | 121,320 | 239,756 | 0.383 | 0.51 | 12,599 | 1858 |
| | 1 | 44 | | | LF | 56,000 | 336,483 | 0.538 | 0.2 | 38,812 | 5815 |
| 10 | 1 | 44 | | 20 | LF | 106,000 | 338,167 | 0.541 | 0.35 | 31,464 | 4580 |
| 15 | | 44 | | 20 | LF | 79,000 | 352,715 | 0.564 | 0.22 | 38,920 | 5671 |
| | | 44 | | | LF | 11,000 | 412,070 | 0.659 | 0 | 57,291 | 8760 |
| 70 | | | 200 | 60 | CC | 460,600 | 564,244 | 0.902 | 1 | | |
| 50 | 2 | | 200 | 40 | CC | 464,600 | 585,473 | 0.936 | 1 | | |

As it is mentioned earlier, the simulation model used for this study is HOMER. This model was run repeatedly using different values for the most important variables and as a result a list of optimal combinations of PV, wind turbine, generator, converter, and battery is provided, which could be implemented as a hybrid system setup. The results are displayed in either of two forms; an overall form in which the top-ranked system configurations are listed according to their net present cost and in a categorized form; where only the least-cost system configuration is considered for each possible system type. Table 3 shows a list of the possible combinations of system components in the overall form. The table has been generated based on inputs selected from the input

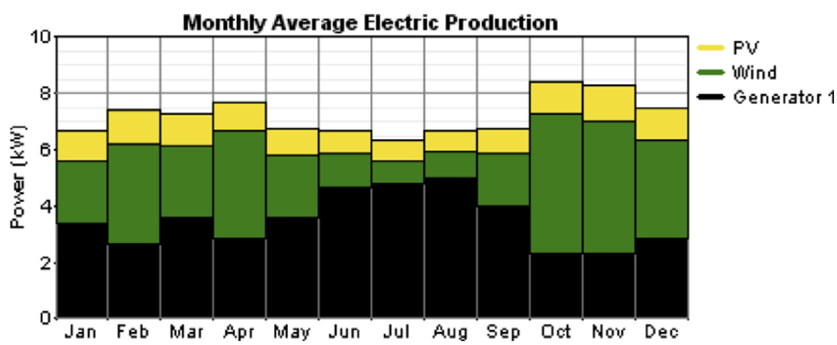
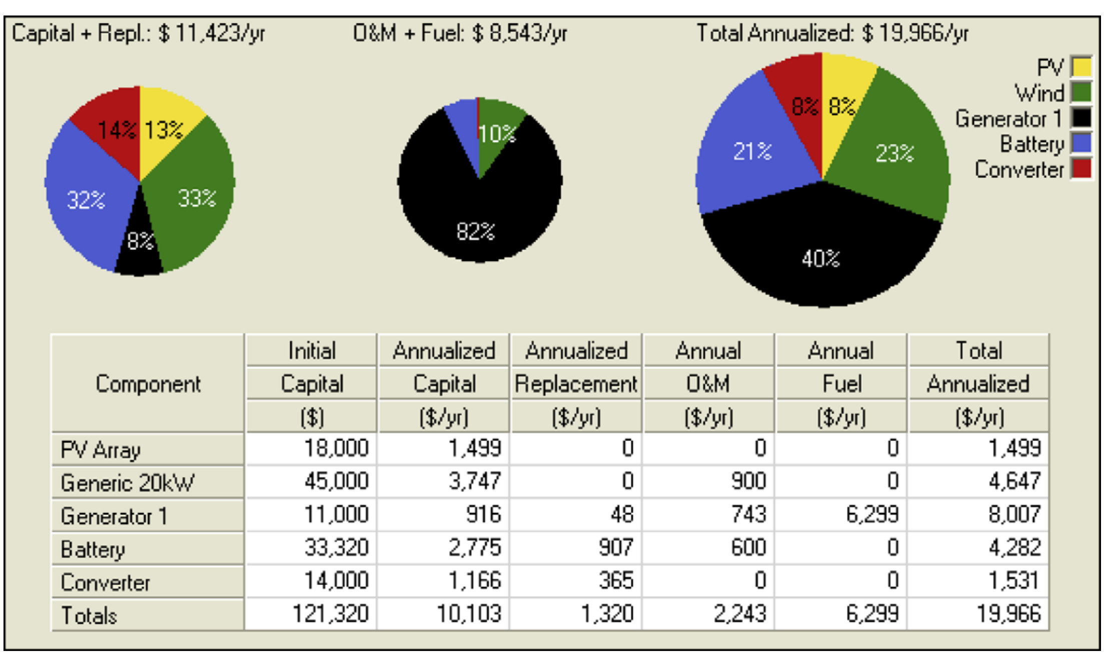



**Fig. 5.** Contribution of the power units for a 51% utilization of renewables.

**Table 5**
System report for 51% renewable resource utilization.

| System architecture | | Sensitivity case | | Annual electric energy production (kW h/year) | | | Annual electric energy consumption (kW h/year) | | | Emissions (kg/year) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PV Array | 5 kW | Solar data | 5.24 kW h/ $m^2$/d | PV array | 8781 | 14% | AC primary load | 50,772 | 97% | $CO_2$ | 33,177 |
| Wind turbine | 1 Generic 20 kW | Wind data | 4.2 m/s | Wind turbine | 23,496 | 37% | Deferrable load | 1306 | 3% | CO | 81.9 |
| Generator | 44 kW | Diesel price | 0.5 $/l | Generator | 30,774 | 49% | Total | 52,077 | 100% | Unburned HC | 9.07 |
| Battery | 40 Surrette 6CS25P | PV capital cost multiplier | 0.6 | Excess electricity | 3860 | 6% | Cost summary | | | Particulate matter | 6.17 |
| Inverter | 20 kW | PV replacement cost multiplier | 0.6 | Unmet load: | 0 | 0% | Total NPC | 239,756 | | $SO_2$ | 66.6 |
| Rectifier | 20 kW | | | Capacity shortage | 0 | 0% | Cost of energy | 0.383 $/kW h | | $NO_x$ | 731 |




| Component | Initial Capital ($) | Annualized Capital ($/yr) | Annualized Replacement ($/yr) | Annual O&M ($/yr) | Annual Fuel ($/yr) | Total Annualized ($/yr) |
|---|---|---|---|---|---|---|
| PV Array | 18,000 | 1,499 | 0 | 0 | 0 | 1,499 |
| Generic 20kW | 45,000 | 3,747 | 0 | 900 | 0 | 4,647 |
| Generator 1 | 11,000 | 916 | 48 | 743 | 6,299 | 8,007 |
| Battery | 33,320 | 2,775 | 907 | 600 | 0 | 4,282 |
| Converter | 14,000 | 1,166 | 365 | 0 | 0 | 1,531 |
| Totals | 121,320 | 10,103 | 1,320 | 2,243 | 6,299 | 19,966 |

**Fig. 6.** Cost summary for a 51% utilization of renewable resources.

summary table (Table 2): 0.5 $/l for diesel price, 0.6 for PV capital multiplier (3600 $/kW). The diesel price is the current price for diesel oil in the country. Interest rates are assumed to be 6.67% and project lifetime is 25 years. With regard to the generator, selected from locally available capacities, 9 kW, 18 kW and 44 kW, the 44 kW generator has been found to be the most cost effective,

As the list is long, part of it has been truncated retaining only those of greatest interest.

Looking at a few of the system setups listed we find the following interesting results. The most cost effective system, i.e., that with the lowest net present cost, is the generator–battery–converter setup, where the generator operates using a cycle charging

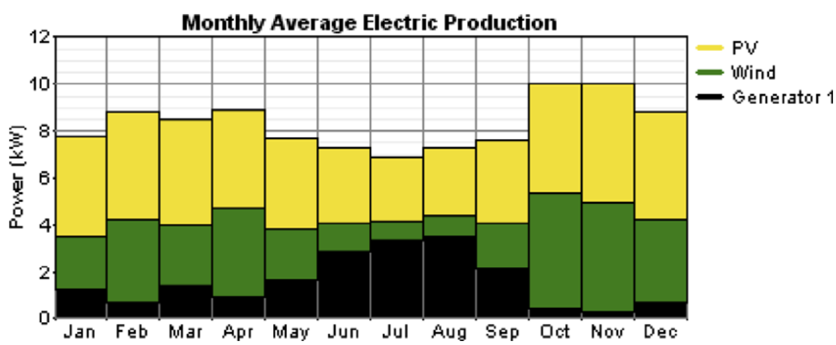
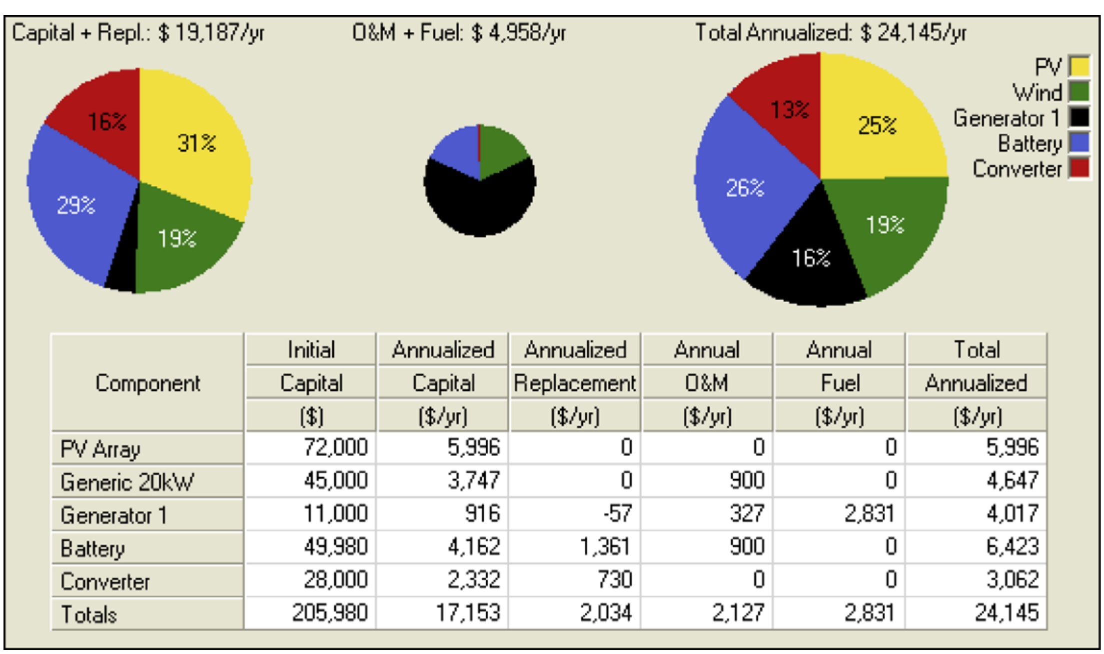



**Fig. 7.** Contribution of the power units at 81% utilization of renewables.

**Table 6**
System report for 81% renewable resource utilization.

| System architecture | | Sensitivity case | | Annual electric energy production (kW h/year) | | | Annual electric energy consumption (kW h/year) | | | Emissions (kg/year) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PV array | 20 kW | Solar data | 5.24 kW h/m$^2$/d | PV array | 35,126 | 48% | AC primary load | 50,772 | 97% | $CO_2$ | 14,909 |
| Wind turbine | 1 Generic 20 kW | Wind data | 4.2 m/s | Wind turbine | 23,496 | 32% | Deferrable load | 1305 | 3% | CO | 36.8 |
| Generator | 44 kW | Diesel price | 0.5 $/l | Generator | 14,019 | 19% | Total | 52,077 | 100% | Unburned HC | 4.08 |
| Battery | 60 Surrette 6CS25P | PV capital cost multiplier | 0.6 | Excess electricity | 7691 | 11% | Cost summary | | | Particulate matter | 2.77 |
| Inverter | 40 kW | PV replacement cost multiplier | 0.6 | Unmet load | 0 | 0% | Total NPC | 289,942 | | $SO_2$ | 29.9 |
| Rectifier | 40 kW | | | Capacity shortage | 0 | 0% | Cost of energy | 0.464 $/kW h | | $NO_x$ | 328 |




| Component | Initial Capital ($) | Annualized Capital ($/yr) | Annualized Replacement ($/yr) | Annual O&M ($/yr) | Annual Fuel ($/yr) | Total Annualized ($/yr) |
|---|---|---|---|---|---|---|
| PV Array | 72,000 | 5,996 | 0 | 0 | 0 | 5,996 |
| Generic 20kW | 45,000 | 3,747 | 0 | 900 | 0 | 4,647 |
| Generator 1 | 11,000 | 916 | -57 | 327 | 2,831 | 4,017 |
| Battery | 49,980 | 4,162 | 1,361 | 900 | 0 | 6,423 |
| Converter | 28,000 | 2,332 | 730 | 0 | 0 | 3,062 |
| Totals | 205,980 | 17,153 | 2,034 | 2,127 | 2,831 | 24,145 |

**Fig. 8.** Cost summary for an 81% utilization of renewable resources.

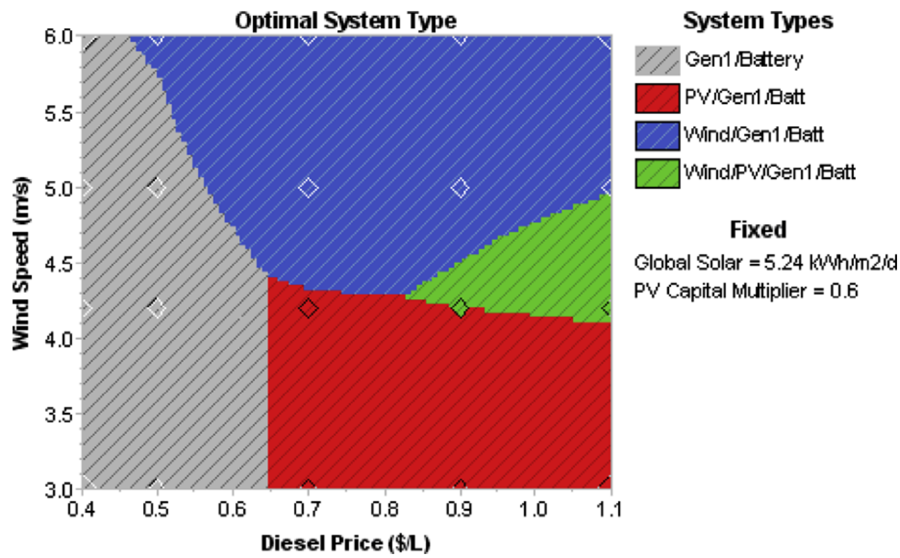



**Fig. 9.** Sensitivity of wind speed to diesel price.

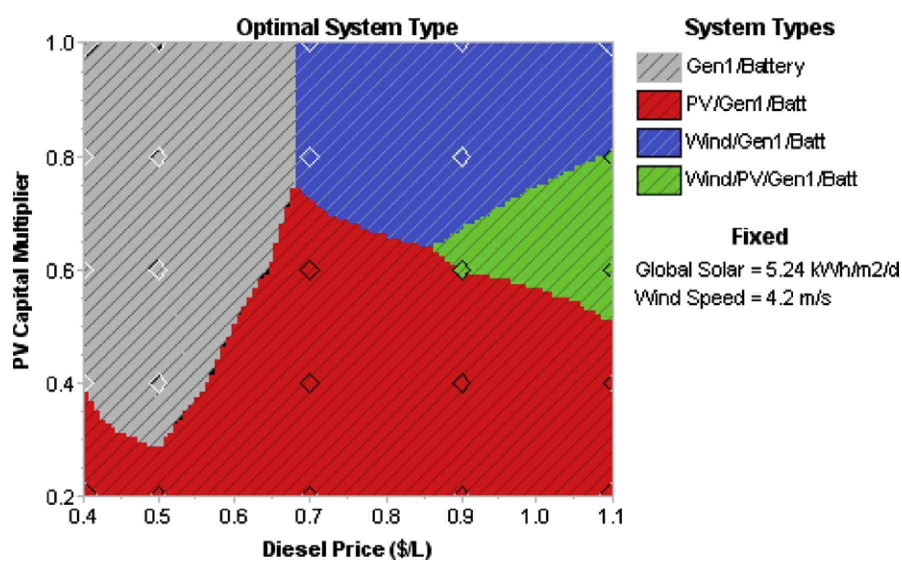



**Fig. 10.** Sensitivity of PV cost to diesel price.

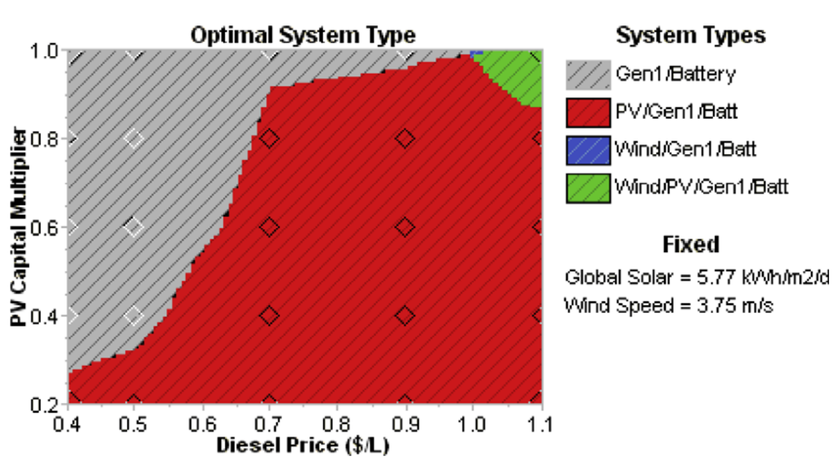



**Fig. 11.** Sensitivity of PV cost to diesel price for Mekele.

(CC) strategy (a dispatch strategy whereby the generator operates at full output power to serve the primary load and any surplus electrical production goes toward the lower-priority objectives). For this setup, the total net present cost (NPC) is $201,609, the cost of energy (COE) is 0.322 $/kW h, there is no contribution from renewable resources, the amount of diesel oil used annually is 18,623 l and the generator operates for 1785 h/year. The advantage of this solution is that the net present cost is the lowest, but renewable resources in no way contribute to the energy supply.

Of those compared, the second most cost effective system is the PV–generator–battery–converter setup, with the generator operating with a Load Following (LF) strategy (a dispatch strategy whereby the generator operates to produce only enough power to meet the primary load; lower-priority objectives such as charging the battery bank or serving the deferrable load is left to the renewable power sources). For this setup the total net present cost (NPC) is $220,728, the cost of energy (COE) is 0.353 $/kW h, the amount of diesel oil used annually is 18,115 l and the generator operates for 2391 h during the year. Again with this scenario, the part contributed by renewable resources is rather small, being only 16%.

Further down in the list, there is a system comprising a PV–wind–generator–battery–converter setup. For this setup the proportion from renewable sources is increased from 16% to 51%, with only a minor increase of 8.5% in the cost. As we can see, the NPC is $239,756 and the COE is 0.383 $/kW h. This could be a good choice for implementation as the contribution made by renewable resources is quite significant. Fig. 5 shows average monthly electrical production and Table 5 gives some important information regarding this system.

The cost breakdown, illustrated by a pie-chart, for this setup, is also given in Fig. 6. In addition, further down in the list, there is another system with 81% utilization of renewable resources. In this setup, the NPC is increased to $289,942 and the COE to 0.464 $/kW h, which is approximately a 21% increase in terms of cost, over the earlier option of 51% utilization. The renewable proportion, however, is increased by about 59%. The average monthly electric production of this setup is given in Fig. 7 and Table 6 also gives some important information regarding this setup.

The cost breakdown, illustrated by a pie-chart, for this setup, is also given in Fig. 8.

The result of optimization, in a categorized form, is given in Table 4; they are ranked according to the NPC but in such a way that the least-cost system is considered for each type of system.

In the table there are two system setups which purely comprise renewable resources (100% contribution). These are PV–battery–converter setup and PV–wind–battery–converter setup. The NPC for the first is $564,244 while that for the second is $585,473 and the COE is 0.902 $/kW h and 0.936 $/kW h, respectively. The

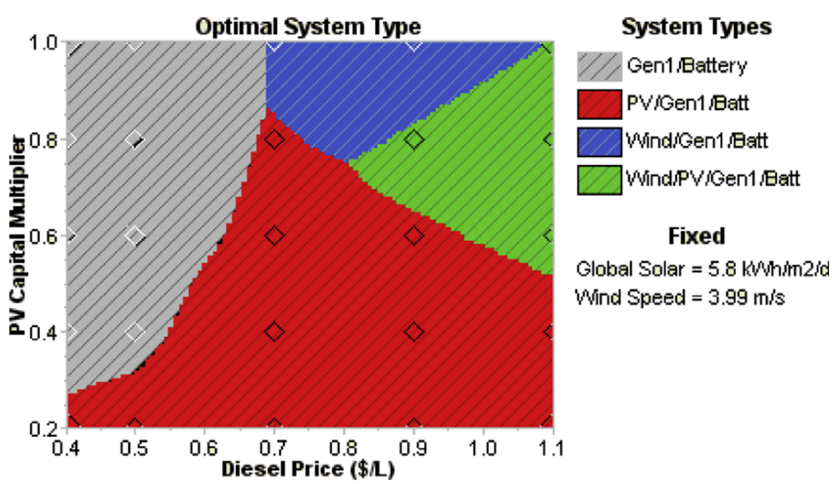



**Fig. 12.** Sensitivity of PV cost to diesel price for Nazret.

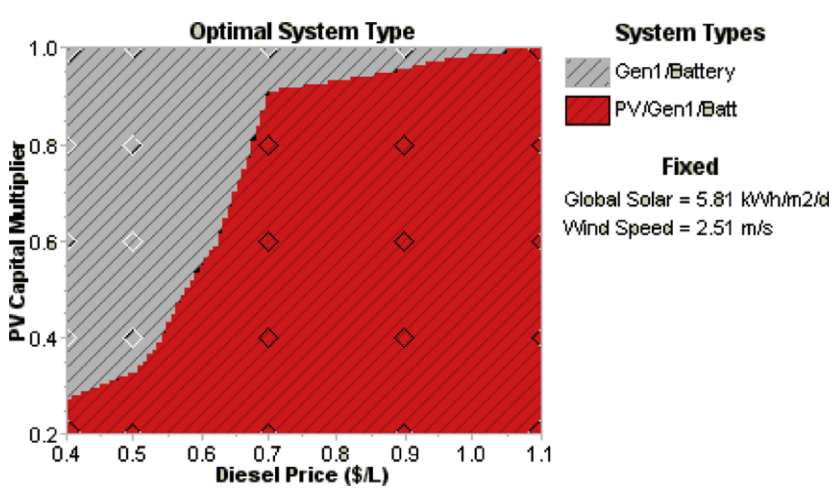



**Fig. 13.** Sensitivity of PV cost to diesel price for Debrezeit.

net present cost for each of these two setups is more than double that of the setup with a 51% renewable utilization. However, the selection of a system to be implemented depends on a number of factors; whether the initial cost is the principal concern or whether there are future benefits to be gained; such as the cleanliness of the energy, a breakaway from fossil fuel price hikes, and also a release from the politics surrounding fossil fuels, etc.

Sensitivity analysis has also been carried out; Figs. 9 and 10 show the respective sensitivities of wind speed and the PV capital cost multiplier to the price of diesel. The current price of diesel oil in the country is approximately 0.5 \$/l and the variation in wind speed is, on average, between 3 and 6 m/s [1].

The current maximum PV price is assumed to be \$6000 $kW^{-1}$ and the minimum \$1200 $kW^{-1}$, assuming a future fall in price.

It should be noted that this study is based on findings which relate to average monthly wind speeds and solar radiation at four typical locations in Ethiopia. However, the results given above are specific to one of the four locations, Addis Ababa and its surroundings.

The complete results for the other three locations are not included within this work, as the size of the article would extend beyond the required limits. However, to give a general overview of the situation, the results obtained from the sensitivity analysis are given for each location in the form of a sensitivity graph. Fig. 11–13 show graphs of PV capital cost multiplier against diesel price for feasible optimal system types at the respective locations.

## 7. Conclusions

Based on the data given in Table 2, the simulation was run and the results show numerous possibilities for implementable setups with different levels of renewable resource utilization. From the list (Table 3), the most cost effective system is the generator–battery–converter setup, with a total net present cost of \$201,609; however, this setup does not include a contribution from renewable resources. Other attractive setups from the list are those with a 51% and 81% utilization of renewable resources. For the setup with a 51% utilization the net present cost is \$239,756 and the levelized cost is 0.383 \$/kW h. This is only a 19% increase in cost but it achieves a 51% use of renewables. For this system there is no unmet demand, no shortage of capacity and excess electricity generation of only 6%. For the setup with 81% utilization of renewable resources the net present cost is \$289,942 and the levelized cost of energy is 0.464 \$/kW h. The cost increase over the 51% utilization is about 20%. Again with this system, there is no unmet demand, no shortage of capacity and only 11% excess electricity. Considering the emission levels of pollutants such as $CO_2$, CO, $SO_2$, and $NO_x$, the latter produces less half that of the earlier option.

As can be observed, the net present cost has to be balanced against the desire to move towards the use of renewable energy. The benefit of such a move cannot be easily expressed in terms of cost. However, the price for diesel oil is increasing over time, while that of PV and the other system components is expected to fall. Under such circumstances, and considering the minor difference in cost suggested by this analysis, it seems realistic to defend the choice of an option which includes renewable energy sources.

Overall, there is little doubt that the hybrid systems studied here, if implemented, would have immense benefits for Ethiopia, a country where the total electricity coverage is less than 15%. At the very least, it would help facilitate the protection of forestry and therefore help prevent soil degradation. This is in addition to an improvement in the quality of life for many and, furthermore, its contribution to a reduction in environmentally polluting emissions. There are also other benefits; free solar and wind energy is utilized, load can be satisfied in an optimal way; the mobilization of investment towards clean energy is facilitated; and most importantly of all, the poor will benefit from the electric light provided.

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